Program overview

19-May-2025 16:39

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Code	Omschrijving	ECTS
Schakelprogramma's ME 2024		
HBO Schakelprogramma's		
Naar Master BME		
Naar Master ME		
IFEEMCS010400	Linear Algebra	5
IFEEMCS012100	Calculus for Engineering, part 1	3
IFEEMCS012200	Calculus for Engineering, part 2	3
IFEEMCS012300	Calculus for Engineering, part 3	3
WB2542	Fluid Mechanics & Heat Transfer	6
WB2543 T1 S	Process Engineering & Thermodynamics	3
WB2630	Advanced Mechanics	6
WB2633 T2	*FEM-practical*	*1*
WB3240-24	Systems and Control Engineering	6
WI1909TH	Differential Equations	3
WI2032TH	Numerical Analysis	3
Naar Master MSE		
IFEEMCS010400	Linear Algebra	5
IFEEMCS012100	Calculus for Engineering, part 1	3
IFEEMCS012200	Calculus for Engineering, part 2	3
IFEEMCS012300	Calculus for Engineering, part 3	3
WB2330	Materials Sciences	6
WB2332	Project Materials Sciences	6
WI1909TH	Differential Equations	3
Naar Master MT		
IFEEMCS010400	Linear Algebra	5
IFEEMCS012100	Calculus for Engineering, part 1	3
IFEEMCS012200	Calculus for Engineering, part 2	3
IFEEMCS012300	Calculus for Engineering, part 3	3
MT2464	Mechanics of Maritime Structures 2	6
MT3463	Ship Motion and Manouvring	6
WB2630	Advanced Mechanics	6
WI1909TH	Differential Equations	3
WI2032TH	Numerical Analysis	3
Supplement HZS		
MT2462	Ship Design	6
Naar Master ODE		
Voor B-CT/CE		
CT1730HBO	Introduction to Geotechnical Engineering	3
CTB1210	Dynamica & Modelvorming	5
CTB2110	Fluid Mechanics	5
CTB2210	Construction Mechanics 3	5
CTB2300	Dynamics of Systems	3
IFEEMCS012100	Calculus for Engineering, part 1	3
IFEEMCS012200	Calculus for Engineering, part 2	3
IFEEMCS012300	Calculus for Engineering, part 3	3
WI1807TH1-21	Linear Algebra	3
WI1909TH	Differential Equations	3
WI2032TH	Numerical Analysis	3
Voor B-MT		
IFEEMCS010400	Linear Algebra	5
IFEEMCS012100	Calculus for Engineering, part 1	3
IFEEMCS012200	Calculus for Engineering, part 2	3
IFEEMCS012300	Calculus for Engineering, part 3	3
MT2464	Mechanics of Maritime Structures 2	6
MT3463	Ship Motion and Manouvring	6
WB2630	Advanced Mechanics	6
WI1909TH	Differential Equations	3
WI2032TH	Numerical Analysis	3
Voor B-WB/LR/TN		
IFEEMCS010400	Linear Algebra	5
IFEEMCS012100	Calculus for Engineering, part 1	3
		3

IFEEMCS012200	Calculus for Engineering, part 2		
IFEEMCS012300	Calculus for Engineering, part 3	3	
WB2542	Fluid Mechanics & Heat Transfer	6	
WB2543 Toets 1	*PE&T tentamen*		*3*
WB2630	Advanced Mechanics	6	
WB2633 T2-S	*FEM-practical*		*1*
WB3240-24	Systems and Control Engineering	6	
WI1909TH	Differential Equations	3	
WI2032TH	Numerical Analysis	3	

Naar Master RO

IFEEMCS010400	Linear Algebra	5
IFEEMCS012100	Calculus for Engineering, part 1	3
IFEEMCS012200	Calculus for Engineering, part 2	3
IFEEMCS012300	Calculus for Engineering, part 3	3
WB2630 T1 S	Rigid-Body Dynamics	3
WB3240-24	Systems and Control Engineering	6
WI1909TH	Differential Equations	3
WI2032TH	Numerical Analysis	3

Naar Master SC

IFEEMCS010400	Linear Algebra	5
IFEEMCS012100	Calculus for Engineering, part 1	3
IFEEMCS012200	Calculus for Engineering, part 2	3
IFEEMCS012300	Calculus for Engineering, part 3	3
WB2235	Signal Analysis	6
WB3240-24	Systems and Control Engineering	6
WI1909TH	Differential Equations	3

WO Schakelprogramma's

Naar Master BME

Naar Master ME

Vanuit B-CT

WB2542 T2 S	Warmteoverdracht	3	
WB2543 Toets 1	*PE&T tentamen*		*3*
WB2630	Advanced Mechanics	6	
WB2633	Advanced Engineering Design Project	6	
WB3240-24	Systems and Control Engineering	6	

Vanuit B-ET

WB2330	Materials Sciences	6
WB2332	Project Materials Sciences	6
WB2542	Fluid Mechanics & Heat Transfer	6
WB2543	Process Engineering & Thermodynamics	6
WB2630	Advanced Mechanics	6
WB2633	Advanced Engineering Design Project	6

Vanuit B-IO

IFEEMCS010400	Linear Algebra	5
WB2330	Materials Sciences	6
WB2332	Project Materials Sciences	6
WB2542	Fluid Mechanics & Heat Transfer	6
WB2543	Process Engineering & Thermodynamics	6
WB2630	Advanced Mechanics	6
WB2633	Advanced Engineering Design Project	6
WB3240-24	Systems and Control Engineering	6
WBMT2048	Mathematics 3	6
WI2032TH	Numerical Analysis	3

Vanuit B-KT

IFEEMCS010400	Linear Algebra	5
WB2330	Materials Sciences	6
WB2542	Fluid Mechanics & Heat Transfer	6
WB2543	Process Engineering & Thermodynamics	6
WB2630	Advanced Mechanics	6
WB2633	Advanced Engineering Design Project	6
WB3240-24	Systems and Control Engineering	6
WBMT2048	Mathematics 3	6
WI2032TH	Numerical Analysis	3

Vanuit B-MST

WB2330	Materials Sciences	6
WB2332	Project Materials Sciences	6
WB2630	Advanced Mechanics	6
WB2633	Advanced Engineering Design Project	6
WB3240-24	Systems and Control Engineering	6
WI2032TH	Numerical Analysis	3

Vanuit B-TN

WB2330	Materials Sciences	6	
WB2543	Process Engineering & Thermodynamics	6	
WB2630 Toets 2	*Continuum Mechanics*		*3*
		6	

WB2633	Advanced Engineering Design Project	
WB3240-24	Systems and Control Engineering	6
Naar Master MSE		
Vanuit B-IO		
IFEEMCS010400	Linear Algebra	5
WB2330	Materials Sciences	6
WB2332	Project Materials Sciences	6
WI1909TH	Differential Equations	3
Vanuit B-KT		
WB2330	Materials Sciences	6
WB2332	Project Materials Sciences	6
Naar Master MT		
Vanuit B-CT		
MT2461	Hydromechanics	6
MT2463	Resistance, Propulsion and Powering 2	6
MT2464	Mechanics of Maritime Structures 2	6
Vanuit B-KT		
MT2461	Hydromechanics	6
MT2462	Ship Design	6
MT2463	Resistance, Propulsion and Powering 2	6
MT2464	Mechanics of Maritime Structures 2	6
MT3463	Ship Motion and Manouvring	6
WB2630	Advanced Mechanics	6
WBMT2048	Mathematics 3	6
WI2032TH	Numerical Analysis	3
Naar Master ODE		
Vanuit B-IO		
IFEEMCS010400	Linear Algebra	5
IFEEMCS012300	Calculus for Engineering, part 3	3
WB2542	Fluid Mechanics & Heat Transfer	6
WB2543	Process Engineering & Thermodynamics	6
WB2630	Advanced Mechanics	6
WB2633 T2-S	*FEM-practical*	*1*
WB3240-24	Systems and Control Engineering	6
WI1909TH	Differential Equations	3
WI2032TH	Numerical Analysis	3
Vanuit B-TBM		
IFEEMCS010400	Linear Algebra	5
IFEEMCS012300	Calculus for Engineering, part 3	3
WB2542	Fluid Mechanics & Heat Transfer	6
WB2543	Process Engineering & Thermodynamics	6
WB2630	Advanced Mechanics	6
WB2633 T2-S	*FEM-practical*	*1*
WB3240-24	Systems and Control Engineering	6
WI1909TH	Differential Equations	3
WI2032TH	Numerical Analysis	3
Vanuit B-TA		
CTB2110	Fluid Mechanics	5
CTB2210	Construction Mechanics 3	5
CTB2300	Dynamics of Systems	3
Naar Master RO		
Vanuit B-KT		
WB2630 T1 S	Rigid-Body Dynamics	3
WB3240-24	Systems and Control Engineering	6
WBMT2048	Mathematics 3	6
WI2032TH	Numerical Analysis	3
Vanuit B-CT		
WB3240-24	Systems and Control Engineering	6
WBMT2048 Toets 1	*Analyse - deeltentamen*	*3*
Vanuit B-CSE		
AESB2210-18	Numerical Mathematics	5
WB2630 T1 S	Rigid-Body Dynamics	3
WBMT2048	Mathematics 3	6
Vanuit B-IO		
IFEEMCS010400	Linear Algebra	5
WB2630 T1 S	Rigid-Body Dynamics	3
WB3240-24	Systems and Control Engineering	6
WBMT2048	Mathematics 3	6
WI2032TH	Numerical Analysis	3
Vanuit B-MST		
WB2630 T1 S	Rigid-Body Dynamics	3
WB3240-24	Systems and Control Engineering	6
WI2032TH	Numerical Analysis	3

Vanuit B-NB

AESB2210-18	Numerical Mathematics	5
WB2630 T1 S	Rigid-Body Dynamics	3
WB3240-24	Systems and Control Engineering	6

Vanuit B-TN

WB2630 T1 S	Rigid-Body Dynamics	3
WB3240-24	Systems and Control Engineering	6

Naar Master SC

Vanuit B-CT, B-AES

WB2235	Signal Analysis	6
WB3240-24	Systems and Control Engineering	6

Vanuit B-TI, B-IO

WB2235	Signal Analysis	6
WB3240-24	Systems and Control Engineering	6
WI1909TH	Differential Equations	3

Naar Master TM

Vanuit Bewegingswetenschappen / Biomedische Wetenschappen

KT2205	Mathematics 3 and Waves	6,5
KT2305	Medical Imaging in Pathologies	6,5
KT2355	Medical Image Processing	3
KT2705	Designing Medical Technology	3
KT2755	An Introduction to the Professional Practice	3,5
KT2905	Skills and Safety 2	4
KT3405	Intensive Care and Computer Simulation	6,5
KT3505	Complex Diagnostics-Therapy-Combinations	6,5

Choose 1 course

KT2505	Senses and Signals	6,5
KT2605	Cardiovascular and Respiratory System and Biomedical Instrumentation 2	6,5
KT2655	Healthcare Information and Organisation	6,5

Vanuit Medische Natuurwetenschappen & Biomedische Technologie

KT2305	Medical Imaging in Pathologies	6,5
KT2355	Medical Image Processing	3
KT2505	Senses and Signals	6,5
KT2605	Cardiovascular and Respiratory System and Biomedical Instrumentation 2	6,5
KT2705	Designing Medical Technology	3
KT2755	An Introduction to the Professional Practice	3,5
KT2905	Skills and Safety 2	4
KT2955	Academic Skills 2	4
KT3405	Intensive Care and Computer Simulation	6,5
KT3505	Complex Diagnostics-Therapy-Combinations	6,5

Vanuit Life, Science and Technology

BM41055	Anatomy and Physiology	4
KT2305	Medical Imaging in Pathologies	6,5
KT2355	Medical Image Processing	3
KT2505	Senses and Signals	6,5
KT2605	Cardiovascular and Respiratory System and Biomedical Instrumentation 2	6,5
KT2705	Designing Medical Technology	3
KT2755	An Introduction to the Professional Practice	3,5
KT2905	Skills and Safety 2	4
KT2955	Academic Skills 2	4
KT3405	Intensive Care and Computer Simulation	6,5
KT3505	Complex Diagnostics-Therapy-Combinations	6,5

Vanuit Geneeskunde

KT1005	Mathematics 1 & 2 and Programming	6
KT2305	Medical Imaging in Pathologies	6,5
KT2355	Medical Image Processing	3
KT2605	Cardiovascular and Respiratory System and Biomedical Instrumentation 2	6,5
KT3405	Intensive Care and Computer Simulation	6,5
KT3505	Complex Diagnostics-Therapy-Combinations	6,5
KT3806	Clinical Technological Final Project	15
KT4001-S	Calculus for Engineering, part 3	1,5
TN2345	Introduction to Waves	3

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Schakelprogramma's ME 2024	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

HBO Schakelprogramma's	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master BME	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl
Introduction 1	See Dutch description

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master ME	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Dr.ir. G.E. Elsinga	
Education Period	1 2	
Start Education	1	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543 T1 S	Process Engineering & Thermodynamics	3
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633 T2	*FEM-practical*	*1*
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master MSE	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2332	Project Materials Sciences	6
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Ir. T.B. Keesom	
Instructor	K.R. Rossi	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Education Method	See Dutch description	
Assessment	See Dutch description	
Enrolment / Application	See Dutch description	
Department	ME Department Materials Science & Engineering	
Judgement	See Dutch description	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master MT	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

MT2464	Mechanics of Maritime Structures 2	6
Responsible Instructor	Dr. C.L. Walters	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Course Contents	Description box (5 ECTS): The course consists of four parts: - Strength of ships: (1) lectures + (2) laboratory exercise - Material science: (3) lectures + (4) laboratory exercise (1) Strength of ships lectures Special topics related to the limit and ultimate strength of ships will be introduced, including buckling (beam and plate), plasticity, fatigue, and fracture.: a) Stress concentration factors b) Buckling of columns: imperfections, eccentricities, Euler, Johnson, Perry-Robertson; c) Buckling of plates: loading, boundary conditions, geometries; d) Effective width of plates: stiffness differences, shear deformations, non-symmetrical/curved cross-sectional forms, post-buckled stress distribution and types of loading; e) Plasticity: form factors, maximum capacity of beams, hinge line theory f) Fracture and fatigue: S-N curves, fatigue crack growth rate, Charpy testing, linear elastic fracture theory, elastic plastic fracture theory. The lecture materials consist of presentations (PowerPoint). The following two reference books are recommended: - Fracture Mechanics: Fundamentals and Applications, ISBN 9781498728133Ship Structural Analysis and Design, ISBN 978-0-939773-78-3 (available for download from TU Delft library)(2) Strength of ships laboratory exercise: plate test (3) Material sciences lectures a) Strengths of materials: characteristics of steel and some non-ferrous ((mechanical) materials, density, structure and properties). b) Strengthening mechanisms on the microstructural scale. c) Thermomechanical processes of metals (deformation, recrystallization, phase transformations). d) Tension test: elastic modulus, yield strength, ultimate tensile strength, failure strain, necking, hardening, and plastic instability. e) Most common welding processes, process descriptions, advantages and disadvantages. f) Influence of welding on materials microstructure, residual stress, deformations, common defects. (4) Material sciences laboratory exercise	
Study Goals	(1) + (2) Strength of ships The recognition of various structural elements (beam and plate elements) of ships with associated supporting conditions and loads. The ability to dimension these structural elements on the basis of strength, stiffness and structural stability. The determination of the stress condition and maximal strength capacity for plate elements loaded in the plane, based on measurements (via digital image correlation). (3) + (4) Materials sciences Insight into the relationship between structure and properties of various materials and into the influence of the production processes on the structure of materials and the use of these materials sciences insights into the design of structures. The ability to describe basic aspects of (fracture-) mechanical material behavior, including the determination of this behavior and the ability to make a simple failure calculation, e.g. for a welded connection. The development of basic knowledge of selected welding processes. Insight into the influence of welding on local microstructures and properties around a welded and the influence of welding on structures. Obtaining insight into the basic principles of melt welding process. Awareness of mechanical tests and the interpretation of results of a tension test.	
Education Method	Lectures and laboratory exercises	
Assessment	a) Laboratory exercise for strength of ships: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. b) Laboratory exercise for material science: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. c) Written exam: this exam consists of a combination of the strength of ships and materials science. 5ECTS	
Department	ME Department Maritime & Transport Technology	

MT3463	Ship Motion and Manouvring	6
Responsible Instructor	Dr.ir. H.J. de Koning Gans	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	ME Department Maritime & Transport Technology	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Supplement HZS

MT2462	Ship Design	6
Responsible Instructor	Dr. A.A. Kana	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Course Contents	This course will focus on the fundamentals of ship design of transport vessels. The classical design spiral and point based design, will be covered as applied to modern transport vessels. Specific focus will also be on the zero-emission energy transition and the influence of alternative fuels on the design of ships, across various aspects, possibly including: ship layout design, operations, regulations, or safety. During the first quarter, half of the effort will be placed on scientific argumentation and communication, with the other half focused on ship design background. The second quarter is focused on ship design. A semester long research assignment examining the influence of alternative energy carriers and converters on the design and operation of a transport vessels will be performed in groups of 2 students. This research assignment will split between a literature study focus in Q1 and a research portion in Q2.	
Study Goals	- Explain and compare primary trade-offs related to alternative power, propulsion, and energy carrier and conversion systems. - Compare and evaluate various aspects of ship design, including: ship systems and ship systems design, and ship layout tradeoffs - Explain the traditional ship design process and its benefits and drawbacks related to sustainable ship design - Search and analyze key challenges and rationale behind sustainable vessel design as part of a literature review - Formulate an argument and communicate key aspects and rationale behind sustainable vessel design in writing as both a literature review and for a specific vessel case study. - Describe and evaluate the influence of environmental maritime considerations on a ship design, as it relates to: ship systems, layout, stability, or ship operations. - Identify and explain primary environmental maritime considerations, which may include aspects of local, national, or international environmental maritime regulations, or vessel retirement recycling	
Education Method	The course will consist of on-campus lectures and a semester long research project in a small group.	
Assessment	Grading will consist of three aspects (1) a group project, (2) scientific argumentation, and (3) an individual assignment	
Department	ME Department Maritime & Transport Technology	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master ODE	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Voor B-CT/CE

CT1730HBO	Introduction to Geotechnical Engineering	3
Responsible Instructor	Prof.dr. M.A. Hicks	
Instructor	Ir. R.A. van der Eijk	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	The course explains the basic concepts of theoretical and applied soil mechanics and geotechnical engineering. Theoretical soil mechanics includes: Soil Characteristics; Groundwater flow; Geomechanics; consolidation, and Shear Strength of Soils. Applied soil mechanics includes: Retaining Structures; Foundations; and Slope Stability.	
Course Contents	Soil Characteristics: Classification of soils; volumetric and gravimetric relationships for soils. Groundwater: Pore pressure and effective stress; Darcys law, permeability and groundwater flow. Geomechanics: Stresses and initial stress state; strains, stress-strain relationships and tangent modulus; Elastic solutions, drained and undrained behaviour. Consolidation: 1D compression, Oedometer test and data interpretation, consolidation coefficient, settlement Shear Strength of Soils: Drained and undrained soil behaviour; total and effective shear strength parameters; Mohr circles; Mohr-Coulomb failure criterion; direct shear test; triaxial test. Retaining Structures: Lateral earth pressure at rest; passive and active pressures Foundations: Geotechnical bearing capacity, shallow foundations Slope Stability: Limit equilibrium methods.	
Study Goals	The main goal of this course is to: Have an understanding of, and be able to apply the following: Soil classification; Groundwater flow; Stresses in the ground; Strains and stiffness; Consolidation; Shear strength of soils; Laboratory testing; Earth retaining structures; Foundations; Slope stability. By the end of the course the student should be able to: Explain fundamentals of soil behaviour in terms of volume changes and shear strength mobilisation. Explain an engineering strategy for analysing geotechnical problems, involving settlements, consolidation, groundwater flow, basic foundations, retaining structures and slope stability. Analyse appropriately and evaluate geotechnical problems, utilising presented theory, on the following topics: elastic behaviour and settlements, consolidation, groundwater flow, basic foundations, retaining structures and slope stability.	
Education Method	Self-study, recorded lectures, tutorials.	
Literature and Study Materials	Materials - Soil mechanics by A. Verruijt, 2001 - Lecture notes - Recorded videos of the course	
Assessment	Computer exam to test the learning objectives of this course.	
Expected prior Knowledge	Basic mechanics, and Mathematics, knowledge of the concept of stress and strain as well as elasticity.	
Academic Skills	Critical thinking, self-studying, note making, time management, application of theory for problems	
Literature & Study Materials	- Soil mechanics by A. Verruijt, 2001 - Craig's Soil Mechanics by R.F. Craig (and J. Knappet), 2012 - Lecture notes	
Judgement	online Final exam (Brightspace based exam) to test the learning objectives of this course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	- Calculator as described in the Examination regulations - Formula sheet	
Collegerama	No	

CTB1210	Dynamica & Modelvorming	5
Responsible Instructor	Dr.ir. A. Gavriilidou	
Instructor	Dr. A. Cabboi	
Instructor	Dr. A.A. Núñez Vicencio	
Instructor	Dr.ir. J. Bosboom	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Collegerama	No	

CTB2110	Fluid Mechanics	5
Responsible Instructor	Dr. C. Chassagne	
Responsible Instructor	Dr.ir. T.S. van den Bremer	
Instructor	Dr. C. Chassagne	
Instructor	Dr.ir. T.S. van den Bremer	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Collegerama	Yes	

CTB2210	Construction Mechanics 3	5
Responsible Instructor	Prof.dr.ir. J.G. Rots	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Collegerama	Yes	

CTB2300	Dynamics of Systems	3
Responsible Instructor	Dr.ir. K.N. van Dalen	
Responsible Instructor	Dr.ir. H. Hendrikse	
Instructor	Dr.ir. H. Hendrikse	
Co-responsible for assignments	Dr.ir. H. Hendrikse	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Dynamics and Modelling Differential Equations for civil engineering	
Parts	The course consists of lectures in which the course content is provided.	
Summary	To be able to model and analyse the dynamic behaviour of several civil engineering systems	
Course Contents	Mechanical system with one degree of freedom, undamped. Formulate equations of motion for free vibration. Forced vibrations for harmonic, exponential, step and impact/pulse loads. Application of initial conditions. Hydraulic systems with one degree of freem without damping; also free and forced motion. Mechanical system with damping; free and forced vibrations. Hydraulic system with damping; free and forced motion. First order systems as a limit case of 2nd-order systems. Mechanical systems with two degrees of freedom, without damping. Formulat equations of motion (mass matrix, stiffness matrix). Determination of eigenfrequencies and eigenperiods. Forced response for harmonic loads.	
Study Goals	Learning Objectives (LO): LO1: To recognize the properties of structural and hydraulic civil engineering systems (for example mass, elasticity, discharges), that are essential for the modelling of such systems. LO2: To be able to formulate simple equations of motion, using the second Newtons Law, with the help of the displacement method. LO3: To be able to analytically derive solutions for formulated equations of motion. LO4: Interpret solutons with respect to the fundamental dynamics parameters, like: eigenfrequency, eigenperiod, free and forced motion, resonance, and steady-state motion. LO5: To be able to predict descent design improvements for dynamic systems that have been studied. 30 hours lectures 3 hours exam 51 hours individual study and exam preparation --------+ 84 hours (3 ECTS)	
Education Method	Lectures, tutorials	
Course Relations	Dynamics Structural mechanics courses	
Literature and Study Materials	Study guide with general information (BrightSpace) Lecture Notes: "Dynamica van Systemen", J. Blaauwendraad, versie 2010. Lecture Slides Other course materials: "Dynamica van Systemen, Vragenstukkenbundel (oefen- en tentamenopgaven)", Studentassistenten CT2022, versie 2011.	
Assessment	Written exam, with open questions	
Permitted Materials during Tests	Calculator and a sheet with main formulas. The latter will be given at the exam together with the exam questions.	
Expected prior Knowledge	Dynamica & Modelforming; Constructuemechanica 1,2,3; Differentiaalvergelijkingen.	
Academic Skills	Analytical thinking; Critical appraisal.	
Literature & Study Materials	Study guide with general information (BrightSpace) Lecture Notes: "Dynamica van Systemen", J. Blaauwendraad, versie 2010. Lecture Slides Other course materials: "Dynamica van Systemen, Vragenstukkenbundel (oefen- en tentamenopgaven)", Studentassistenten CT2022, versie 2011.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator and a sheet with main formulas. The latter will be given at the exam together with the exam questions.	
Collegerama	Yes	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WI1807TH1-21	Linear Algebra	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programme students are applying for.	
Course Contents	- Systems of linear equations, elementary row-operations, (reduced) echelon form - Vector equations, spans - Solution sets of (non)homogeneous systems of linear equations - Linear independence - Linear transformations and standard matrices - Matrix algebra (sum, product), inverse of a matrix - Subspaces in R^n, column space and null space - Basis and Dimension - Determinants - Eigenvalues and eigenvectors - Diagonalization	
Study Goals	After successfully completing the course you will be able to: - solve systems of linear equations. - identify if a system of linear equations is solvable, and if the solution is unique. - show the equivalence between a vector equation, a matrix equation and a system of linear equations. - give a geometric description of solution sets of vector equations. - determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - identify if a transformation is linear and determine the standard matrix of a linear transformation. - perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - determine the determinant of a matrix by applying the properties of a determinant. - apply the properties of determinants (product, transpose, inverse). - determine the eigenvalues and eigenvectors of a matrix. - determine the algebraic and geometric multiplicity of the eigenvalues of a matrix. - determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - analyze the case of complex eigenvalues and eigenvectors.	
Education Method	Lectures, 4 hours per week (2x2)	
Literature and Study Materials	- Book: David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and its Applications, Sixth edition, Global Edition, ISBN-13: 978-1292351216. - Slides on Brightspace. - Prelecture videos. - Online homework assignments (Grasple). - Old exams on Brightspace.	
Books	David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and its Applications, Sixth edition, Global Edition, ISBN-13: 978-1292351216.	
Assessment	Digital exam (in Grasple) on campus.	
Remarks	Lectures will be in Dutch! Exams will be in offered in Dutch and also in English.	
Co-Instructor	Dr. E. Emsiz	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Voor B-MT

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

MT2464	Mechanics of Maritime Structures 2	6

Responsible Instructor Dr. C.L. Walters

Education Period 3

Start Education 3

Exam Period 3
4

Course Language Dutch
English

Course Contents

Description box (5 ECTS):
The course consists of four parts:
- Strength of ships: (1) lectures + (2) laboratory exercise
- Material science: (3) lectures + (4) laboratory exercise
(1) Strength of ships lectures
Special topics related to the limit and ultimate strength of ships will be introduced, including buckling (beam and plate), plasticity, fatigue, and fracture.:
a) Stress concentration factors
b) Buckling of columns: imperfections, eccentricities, Euler, Johnson, Perry-Robertson;
c) Buckling of plates: loading, boundary conditions, geometries;
d) Effective width of plates: stiffness differences, shear deformations, non-symmetrical/curved cross-sectional forms, post-buckled stress distribution and types of loading;
e) Plasticity: form factors, maximum capacity of beams, hinge line theory
f) Fracture and fatigue: S-N curves, fatigue crack growth rate, Charpy testing, linear elastic fracture theory, elastic plastic fracture theory.
The lecture materials consist of presentations (PowerPoint). The following two reference books are recommended:
- Fracture Mechanics: Fundamentals and Applications, ISBN 9781498728133Ship Structural Analysis and Design, ISBN 978-0-939773-78-3 (available for download from TU Delft library)(2) Strength of ships laboratory exercise: plate test
(3) Material sciences lectures
a) Strengths of materials: characteristics of steel and some non-ferrous ((mechanical) materials, density, structure and properties).
b) Strengthening mechanisms on the microstructural scale.
c) Thermomechanical processes of metals (deformation, recrystallization, phase transformations).
d) Tension test: elastic modulus, yield strength, ultimate tensile strength, failure strain, necking, hardening, and plastic instability.
e) Most common welding processes, process descriptions, advantages and disadvantages.
f) Influence of welding on materials microstructure, residual stress, deformations, common defects.
(4) Material sciences laboratory exercise

Study Goals

(1) + (2) Strength of ships
The recognition of various structural elements (beam and plate elements) of ships with associated supporting conditions and loads.

The ability to dimension these structural elements on the basis of strength, stiffness and structural stability. The determination of the stress condition and maximal strength capacity for plate elements loaded in the plane, based on measurements (via digital image correlation).

(3) + (4) Materials sciences
Insight into the relationship between structure and properties of various materials and into the influence of the production processes on the structure of materials and the use of these materials sciences insights into the design of structures.

The ability to describe basic aspects of (fracture-) mechanical material behavior, including the determination of this behavior and the ability to make a simple failure calculation, e.g. for a welded connection.

The development of basic knowledge of selected welding processes. Insight into the influence of welding on local microstructures and properties around a welded and the influence of welding on structures. Obtaining insight into the basic principles of melt welding process. Awareness of mechanical tests and the interpretation of results of a tension test.

Education Method Lectures and laboratory exercises

Assessment

a) Laboratory exercise for strength of ships: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS.
b) Laboratory exercise for material science: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS.
c) Written exam: this exam consists of a combination of the strength of ships and materials science. 5ECTS

Department ME Department Maritime & Transport Technology

MT3463	Ship Motion and Manouvring	6

Responsible Instructor Dr.ir. H.J. de Koning Gans

Education Period 3

Start Education 3

Exam Period 3
4

Course Language Dutch

Department ME Department Maritime & Transport Technology

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Voor B-WB/LR/TN

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Dr.ir. G.E. Elsinga	
Education Period	1 2	
Start Education	1	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543 Toets 1	*PE&T tentamen*	*3*
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633 T2-S	*FEM-practical*	*1*
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master RO	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WB2630 T1 S	Rigid-Body Dynamics	3
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Ir. P.H. de Jong	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Study Goals	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Education Method	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Assessment	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master SC

Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl
Introduction 1	Studenten met een diploma HBO-Werktuigbouwkunde, Luchtvaarttechniek, Mechatronica, Elektrotechniek, Technische Wiskunde of Toegepaste Natuurkunde worden toegelaten tot dit schakelprogramma. Toelating tot de Masteropleiding Systems & Control vindt uitsluitend plaats als het schakelprogramma volledig is afgerond. Schakelvakken worden in het Nederlands gegeven. Eén van de hieronder genoemde vakken is verplicht: CTB2200 - Statistiek WI2031TH - Kansrekening & Statistiek

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WB2235	Signal Analysis	6
Responsible Instructor	Dr.ing. R. van de Plas	
Instructor	G. de Albuquerque Gleizer	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	WI2030WBMT Wiskunde 3 (Differentiaalvergelijkingen) WI2031WBMT Wiskunde 4 (Kansrekening en Statistiek) should be heard in parallel to the course	
Course Contents	This course treats the analysis, filtering and detection of signals influenced by linear time-invariant systems as they occur in typical mechanical and mechatronic applications. Signals will be considered in both deterministic and stochastic formulations. Special emphasis is put on discrete-time techniques that can be easily implemented in practice. The main topics of the course are: 1. Elementary properties of signals and systems (even, odd, linear, stable, causal, linear-phase, group-delay, all-pass, minimum-phase) 2. Fourier and Hilbert transform, one and two-sided Laplace and Z-transforms 3. Bode plots and Bodes phase-gain relationship 4. Continuous-time low-pass filters 5. Modulation and demodulation 6. Analog-to-digital and digital-to-analog conversion 7. Design of discrete-time filters using windowing and impulse invariance 8. Elementary properties of random processes (i.i.d., w.s.s., joint w.s.s., independence); key indicators such as mean, auto and cross-correlation/covariance, and spectral densities 9. Identification of linear time-invariant discrete-time systems in the frequency domain 10. Discrete-time Wiener filtering (finite impulse response and non-causal) 11. Detection of discrete-time signals in Gaussian noise	
Study Goals	Students can 1. Evaluate if a signal or system possesses an elementary property; utilize these in simple problems; give examples that exhibit desired elementary properties 2. Compute Fourier, Laplace, z and Hilbert transforms, both directly and using transform pairs and properties 3. Sketch Bode plots for rational transfer functions 4. Determine the specifications of continuous-time low-pass filters from their frequency response; design low-pass filters that fulfill given specifications 5. Analyze typical modulation and demodulation schemes; add missing components 6. Evaluate the impact of A/D and D/A conversion schemes; choose sampling intervals 7. Design discrete-time filters using impulse invariance, windowing, and Wiener techniques 8. Evaluate if a random process possesses an elementary property; utilize them in simple problems; determine key indicators of random processes 9. Identify the frequency response of a linear time-invariant discrete-time system from actual measurements; specify the bias and the variance 10. Derive optimal detection rules for a known signal in Gaussian noise	
Education Method	Lectures, problem sets and practice sessions (werkcolleges).	
Literature and Study Materials	- A.V. Oppenheim, A.S. Willsky and S. Hamid, Signals and Systems, Pearson New International Edition, 2nd ed, 2013. - A.V. Oppenheim and G. Verghese, Signals, Systems and Inference, Class Notes for 6.011: Introduction to Communication, Control and Signal Processing, MIT, Spring 2010. Available online: http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-011-introduction-to-communication-control-and-signal-processing-spring-2010/readings/ - For some parts of the lecture, external materials will be used. These materials will either be available online from within the network of TU Delft, or be provided via Brightspace.	
Assessment	Written exam. Students will only be allowed to bring a non-programmable and non-graphing calculator. A formula sheet will be provided.	
Department	ME Department Delft Center for Systems and Control	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

WO Schakelprogramma's	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master BME	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master ME	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-CT

WB2542 T2 S	Warmteoverdracht	3
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Dr.ir. M.J. Tummers	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543 Toets 1	*PE&T tentamen*	*3*
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633	Advanced Engineering Design Project	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Course Language	Dutch	
Course Contents	wb2633	
Study Goals	see section descriptions	
Education Method	see section descriptions	
Assessment	see section descriptions	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-ET

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2332	Project Materials Sciences	6
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Ir. T.B. Keesom	
Instructor	K.R. Rossi	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Education Method	See Dutch description	
Assessment	See Dutch description	
Enrolment / Application	See Dutch description	
Department	ME Department Materials Science & Engineering	
Judgement	See Dutch description	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Dr.ir. G.E. Elsinga	
Education Period	1 2	
Start Education	1	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543	Process Engineering & Thermodynamics	6
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Instructor	Dr. R. Delfos	
Responsible for assignments	L. Botto	
Co-responsible for assignments	Dr.ir. T.M.J. Nijssen	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Modelling Practicals Process Project Small groups Sustainability Technology Thermodynamics Transport phenomena Water Engineering exergy	
Department	ME Department Process & Energy	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633	Advanced Engineering Design Project	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Course Language	Dutch	
Course Contents	wb2633	
Study Goals	see section descriptions	
Education Method	see section descriptions	
Assessment	see section descriptions	
Department	ME Department Precision & Microsystems Engineering	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-IO

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2332	Project Materials Sciences	6
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Ir. T.B. Keesom	
Instructor	K.R. Rossi	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Education Method	See Dutch description	
Assessment	See Dutch description	
Enrolment / Application	See Dutch description	
Department	ME Department Materials Science & Engineering	
Judgement	See Dutch description	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Dr.ir. G.E. Elsinga	
Education Period	1 2	
Start Education	1	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543	Process Engineering & Thermodynamics	6
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Instructor	Dr. R. Delfos	
Responsible for assignments	L. Botto	
Co-responsible for assignments	Dr.ir. T.M.J. Nijssen	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Modelling Practicals Process Project Small groups Sustainability Technology Thermodynamics Transport phenomena Water Engineering exergy	
Department	ME Department Process & Energy	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633	Advanced Engineering Design Project	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Course Language	Dutch	
Course Contents	wb2633	
Study Goals	see section descriptions	
Education Method	see section descriptions	
Assessment	see section descriptions	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses.	
Course Contents	WBMT2048 T1 - Analysis 3: - Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss. WBMT2048 T2 - Differential Equations: - Linear first and second order equations; - Laplace transform; - Systems of linear equations - Autonomous systems and stability; - Fourier series; - Partial differential equations.	
Study Goals	Study goals for each part can be found on the corresponding StudyGuide-page (WBMT2048 Toets 1 or WBMT2048 Toets 2).	
Education Method	Lectures 4 hours per week.	
Literature and Study Materials	WBMT2048 Toets 1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that will also be made available via Brightspace. WBMT2048 Toets T2: William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	WBMT2048 T1: - Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the entire course, which lasts 180 minutes. It is not possible to resit only one partial exam. WBMT2048 T2: - Written exams (see Brightspace for more details).	
Exam Hours	WBMT2048 T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048 T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048 T1: A formula sheet (see Brightspace). NO calculators. WBMT2048 T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	
Contact	WBMT2048@tudelft.nl	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-KT

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Dr.ir. G.E. Elsinga	
Education Period	1 2	
Start Education	1	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543	Process Engineering & Thermodynamics	6
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Instructor	Dr. R. Delfos	
Responsible for assignments	L. Botto	
Co-responsible for assignments	Dr.ir. T.M.J. Nijssen	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Modelling Practicals Process Project Small groups Sustainability Technology Thermodynamics Transport phenomena Water Engineering exergy	
Department	ME Department Process & Energy	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633	Advanced Engineering Design Project	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Course Language	Dutch	
Course Contents	wb2633	
Study Goals	see section descriptions	
Education Method	see section descriptions	
Assessment	see section descriptions	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WBMT2048	Mathematics 3	6

Responsible Instructor	Dr.ir. R.F. Swarttouw
Education Period	1 2
Start Education	1 2
Exam Period	Different, to be announced
Course Language	Dutch
Required for	(Fluid) Mechanics, Mechatronics.
Expected prior knowledge	First year Mathematics courses.
Course Contents	WBMT2048 T1 - Analysis 3: - Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss. WBMT2048 T2 - Differential Equations: - Linear first and second order equations; - Laplace transform; - Systems of linear equations - Autonomous systems and stability; - Fourier series; - Partial differential equations.
Study Goals	Study goals for each part can be found on the corresponding StudyGuide-page (WBMT2048 Toets 1 or WBMT2048 Toets 2).
Education Method	Lectures 4 hours per week.
Literature and Study Materials	WBMT2048 Toets 1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that will also be made available via Brightspace. WBMT2048 Toets T2: William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.
Assessment	WBMT2048 T1: - Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the entire course, which lasts 180 minutes. It is not possible to resit only one partial exam. WBMT2048 T2: - Written exams (see Brightspace for more details).
Exam Hours	WBMT2048 T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048 T2: 180 minutes.
Permitted Materials during Tests	WBMT2048 T1: A formula sheet (see Brightspace). NO calculators. WBMT2048 T2: A table of Laplace transforms (see Brightspace). NO calculators.
Enrolment / Application	See Dutch comment.
Judgement	See Dutch comment.
Contact	WBMT2048@tudelft.nl

WI2032TH	Numerical Analysis	3

Responsible Instructor	Dr. N.V. Budko
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	Dutch
Course Contents	See WBMT2049 T2
Study Goals	See WBMT2049 T2
Education Method	See WBMT2049 T2
Assessment	See WBMT2049 T2 and WBMT2049 T3
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.
Co-Instructor	D. Toshniwal

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-MST

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2332	Project Materials Sciences	6
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Ir. T.B. Keesom	
Instructor	K.R. Rossi	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Education Method	See Dutch description	
Assessment	See Dutch description	
Enrolment / Application	See Dutch description	
Department	ME Department Materials Science & Engineering	
Judgement	See Dutch description	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633	Advanced Engineering Design Project	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Course Language	Dutch	
Course Contents	wb2633	
Study Goals	see section descriptions	
Education Method	see section descriptions	
Assessment	see section descriptions	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-TN

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2543	Process Engineering & Thermodynamics	6
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Instructor	Dr. R. Delfos	
Responsible for assignments	L. Botto	
Co-responsible for assignments	Dr.ir. T.M.J. Nijssen	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Modelling Practicals Process Project Small groups Sustainability Technology Thermodynamics Transport phenomena Water Engineering exergy	
Department	ME Department Process & Energy	

WB2630 Toets 2	Continuum Mechanics	3

Responsible Instructor	Dr.ir. L.F.P. Noel
Contact Hours / Week x/x/x/x	lectures 0/2/0/0 Practicals 0/2/0/0 Office houres 0/2/0/0
Education Period	2
Start Education	2
Exam Period	2 3
Course Language	English
Course Contents	Linear 3D continuum theory 1. Kinematics equations in 3D 2. Deformation tensor in 3D, linear and nonlinear 3. Stretch in arbitrary direction 4. Equilibrium equations in 3D 5. Stress vector and tensor in 3D, linear 6. Principal stresses and directions, normal and shear stress acting on a plane 7. Constitutive equations in 3D 8. Generalized Hookes law 9. Plane stress/strain state Energy principles 1. Principle of virtual work (PVW) without/with deformations 2. Principle of minimum potential energy (MPE) 3. Approximate solutions based on MPE Discrete linear systems 1. Systematic way of solving problems with n degrees of freedom (DOF) 2. Analysis of discrete linear systems (spring, truss) DOF, generalized deformations, generalized stresses, generalized constitutive equations, discrete continuity and equilibrium, member/global stiffness, assembly, boundary conditions, solution Finite Element Method 1. Approximations based on MPE or PVW 2. Derivation of simple elements and isoparametric elements 3. Analysis of structures with the FEM DOF, element/global stiffness, assembly, boundary conditions, distributed loads, solution 4. Discussion on element choice, mesh refinement, convergence
Study Goals	By the end of the course, the student is able to: Linear 3D continuum theory 1. Recognize situations where the continuum theory is applicable 2. Derive kinematics equations in 3D a. Derive relation between displacements and deformations b. Provide interpretation for the components of the strain tensor c. Recognize situations where kinematics equations are applicable 3. Use kinematics equations a. Determine deformations, a strain tensor for known displacements b. Determine deformations, stretches in specified directions c. Explain the relevance of and the procedure to measure deformations in specified directions 4. Derive the equilibrium equations in 3D a. Explain equilibrium equations in 3D b. Explain the relations between external forces, internal forces, and stresses c. Recognize situations where equilibrium equations are applicable 5. Use equilibrium equations a. Determine a stress vector, a stress tensor b. Determine a normal stress, a shear stress c. Determine principal stresses and directions 6. Derive generalized Hookes law in 3D a. Describe its relevance and limitations 7. Apply generalized Hookes law 8. Identify plane stress or plane strain states Energy principles 1. Describe the principle of virtual work (PVW) 2. Derive equilibrium equations using the PVW 3. Recognize situations where the PVW is applicable 4. Describe the principle of minimum potential energy (MPE) 5. Derive equilibrium equations using the MPE 6. Recognize situations where the MPE is applicable 7. Build approximate solutions using the MPE Discrete linear systems 1. Build discrete linear systems for given structures 2. Analyze structures with discrete linear systems a. Build local member stiffness matrices b. Build global system stiffness matrices c. Set up system of equations for discrete linear systems 3. Solve problems using discrete linear systems a. Impose boundary conditions b. Solve for displacements at joints c. Post-process for reaction forces, stresses, and strains Finite Element Method 1. Describe the basic concepts of the Finite Element Method (FEM) 2. Derive simple and isoparametric elements based on the PVW or on the MPE a. Build a geometry and a displacement approximation over an element b. Build the stiffness matrix for an element c. Build energy conjugated loads for an element d. Perform numerical integration over an element 3. Analyze structures with the FEM a. Build local element stiffness matrices b. Build global structure stiffness matrices

	c. Set up systems of equations for FEM model 4. Solve problems with the FEM a. Compute energy equivalent nodal forces b. Impose boundary conditions c. Solve for displacements at nodes d. Post-process for reaction forces, stresses, and strains 5. Identify appropriate element choice, mesh density, mesh refinement, convergence requirements
Education Method	Contact hours Plenary sessions, including lectures and formative in-class tests (4 hours per week) Questions hours in groups, including practice and exercises (2 hours per weeks) Formative in-class tests Formative in-class tests are organized every week right after the lectures and may lead to a bonus on the final course grade (maximum +1.0 point on the final grade). Bonus credits are only valid for the exam, not for the resit. During the in-class tests, you may collaborate with your fellow students and use the lecture slides. The use of any type of electronic communication (smart phone, texting app, email, social media, etc.) or calculator is considered fraud and will be treated as such. Grade calculation The final grade G is computed based on the exam grade GE and the bonus B(GT) obtained from the test grade GT or based on the resit grade GR only. Provided the exam grade GE is 5.00 or higher, G = GE + B(GT). Otherwise, G = GE. For the resit, the final grade is solely based on the resit grade GR and G = GR. The test grade GT can lead to a bonus B(GT) on the final grade and is based on the average of the best 75% of individual test grades. Missing a test leads to a grade of 0.0 for that particular test. Study materials Lecture slides, exercises instructions, previous exams provided on Brightspace
Assessment	Written exam. Taking part in formative in-class tests can lead to bonus on the final grade.
Department	ME Department Precision & Microsystems Engineering
Contact	continuum-mechanics-ME@tudelft.nl

WB2633	**Advanced Engineering Design Project**	**6**
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Course Language	Dutch	
Course Contents	wb2633	
Study Goals	see section descriptions	
Education Method	see section descriptions	
Assessment	see section descriptions	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	**Systems and Control Engineering**	**6**
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master MSE	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-IO

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2332	Project Materials Sciences	6
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Ir. T.B. Keesom	
Instructor	K.R. Rossi	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Education Method	See Dutch description	
Assessment	See Dutch description	
Enrolment / Application	See Dutch description	
Department	ME Department Materials Science & Engineering	
Judgement	See Dutch description	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-KT

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2332	Project Materials Sciences	6
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Ir. T.B. Keesom	
Instructor	K.R. Rossi	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Education Method	See Dutch description	
Assessment	See Dutch description	
Enrolment / Application	See Dutch description	
Department	ME Department Materials Science & Engineering	
Judgement	See Dutch description	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master MT

Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-CT

MT2461	Hydromechanics	6
Responsible Instructor	Prof. G.D. Weymouth	
Instructor	G. Jacobi	
Instructor	B. Font	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	How does the water move around a ship and in waves, and what forces and moments result? You learn all about this in Hydrodynamics, deepening your understanding from WVAI and preparing you for WVA II and Ship Maneuvering and Seakeeping. You start from the basics of fluid motion and momentum and build up to flows around submerged bodies such as underwater vehicles, control surfaces like rudders, and lifting surfaces including hydrofoils. You'll learn how fluid forces are generated on these bodies, and to predict those forces using hand calculations, experiments and computational methods. Waves are also a critical part of hydrodynamics, and you will learn how they are formed and travel across the ocean surface. We'll discuss how large ships create these waves and the power required to push them through the water.	
Study Goals	After successfully completing this course, you will be able to: - Understand fundamental concepts of Conservation of fluid volume, mass, and momentum applied to control volumes. - Carry out calculations of potential flow past immersed bodies applied to dynamic lift like sails, hydrofoils and planning. - Carry out calculations of viscous flow in boundary layers and wakes applied to hull forms and roll damping. - Predict ocean wave speed and amplitude and understand their relevance for wave forces. - Understand the limitations of a model with respect to its application and deviations from reality.	
Education Method	Lectures & practicals, both mandatory	
Assessment	- 25% online weekly quizzes, 50% cumulative minimum - 25% lab report, 50% minimum. - 50% written exam, 50% minimum	
Department	ME Department Maritime & Transport Technology	

MT2463	Resistance, Propulsion and Powering 2	6
Responsible Instructor	Dr.ir. P. de Vos	
Instructor	E.H.M. Ulijn	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	see Dutch version	
Course Contents	see Dutch version	
Study Goals	see Dutch version	
Education Method	see Dutch version	
Assessment	see Dutch version	
Department	ME Department Maritime & Transport Technology	

MT2464	Mechanics of Maritime Structures 2	6
Responsible Instructor	Dr. C.L. Walters	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Course Contents	Description box (5 ECTS): The course consists of four parts: - Strength of ships: (1) lectures + (2) laboratory exercise - Material science: (3) lectures + (4) laboratory exercise (1) Strength of ships lectures Special topics related to the limit and ultimate strength of ships will be introduced, including buckling (beam and plate), plasticity, fatigue, and fracture.: a) Stress concentration factors b) Buckling of columns: imperfections, eccentricities, Euler, Johnson, Perry-Robertson; c) Buckling of plates: loading, boundary conditions, geometries; d) Effective width of plates: stiffness differences, shear deformations, non-symmetrical/curved cross-sectional forms, post-buckled stress distribution and types of loading; e) Plasticity: form factors, maximum capacity of beams, hinge line theory f) Fracture and fatigue: S-N curves, fatigue crack growth rate, Charpy testing, linear elastic fracture theory, elastic plastic fracture theory. The lecture materials consist of presentations (PowerPoint). The following two reference books are recommended: - Fracture Mechanics: Fundamentals and Applications, ISBN 9781498728133Ship Structural Analysis and Design, ISBN 978-0-939773-78-3 (available for download from TU Delft library)(2) Strength of ships laboratory exercise: plate test (3) Material sciences lectures a) Strengths of materials: characteristics of steel and some non-ferrous ((mechanical) materials, density, structure and properties). b) Strengthening mechanisms on the microstructural scale. c) Thermomechanical processes of metals (deformation, recrystallization, phase transformations). d) Tension test: elastic modulus, yield strength, ultimate tensile strength, failure strain, necking, hardening, and plastic instability. e) Most common welding processes, process descriptions, advantages and disadvantages. f) Influence of welding on materials microstructure, residual stress, deformations, common defects. (4) Material sciences laboratory exercise	
Study Goals	(1) + (2) Strength of ships The recognition of various structural elements (beam and plate elements) of ships with associated supporting conditions and loads. The ability to dimension these structural elements on the basis of strength, stiffness and structural stability. The determination of the stress condition and maximal strength capacity for plate elements loaded in the plane, based on measurements (via digital image correlation). (3) + (4) Materials sciences Insight into the relationship between structure and properties of various materials and into the influence of the production processes on the structure of materials and the use of these materials sciences insights into the design of structures. The ability to describe basic aspects of (fracture-) mechanical material behavior, including the determination of this behavior and the ability to make a simple failure calculation, e.g. for a welded connection. The development of basic knowledge of selected welding processes. Insight into the influence of welding on local microstructures and properties around a welded and the influence of welding on structures. Obtaining insight into the basic principles of melt welding process. Awareness of mechanical tests and the interpretation of results of a tension test.	
Education Method	Lectures and laboratory exercises	
Assessment	a) Laboratory exercise for strength of ships: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. b) Laboratory exercise for material science: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. c) Written exam: this exam consists of a combination of the strength of ships and materials science. 5ECTS	
Department	ME Department Maritime & Transport Technology	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-KT

MT2461	Hydromechanics	6
Responsible Instructor	Prof. G.D. Weymouth	
Instructor	G. Jacobi	
Instructor	B. Font	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	How does the water move around a ship and in waves, and what forces and moments result? You learn all about this in Hydrodynamics, deepening your understanding from WVAI and preparing you for WVA II and Ship Maneuvering and Seakeeping. You start from the basics of fluid motion and momentum and build up to flows around submerged bodies such as underwater vehicles, control surfaces like rudders, and lifting surfaces including hydrofoils. You'll learn how fluid forces are generated on these bodies, and to predict those forces using hand calculations, experiments and computational methods. Waves are also a critical part of hydrodynamics, and you will learn how they are formed and travel across the ocean surface. We'll discuss how large ships create these waves and the power required to push them through the water.	
Study Goals	After successfully completing this course, you will be able to: - Understand fundamental concepts of Conservation of fluid volume, mass, and momentum applied to control volumes. - Carry out calculations of potential flow past immersed bodies applied to dynamic lift like sails, hydrofoils and planning. - Carry out calculations of viscous flow in boundary layers and wakes applied to hull forms and roll damping. - Predict ocean wave speed and amplitude and understand their relevance for wave forces. - Understand the limitations of a model with respect to its application and deviations from reality.	
Education Method	Lectures & practicals, both mandatory	
Assessment	- 25% online weekly quizzes, 50% cumulative minimum - 25% lab report, 50% minimum. - 50% written exam, 50% minimum	
Department	ME Department Maritime & Transport Technology	

MT2462	Ship Design	6
Responsible Instructor	Dr. A.A. Kana	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Course Contents	This course will focus on the fundamentals of ship design of transport vessels. The classical design spiral and point based design, will be covered as applied to modern transport vessels. Specific focus will also be on the zero-emission energy transition and the influence of alternative fuels on the design of ships, across various aspects, possibly including: ship layout design, operations, regulations, or safety. During the first quarter, half of the effort will be placed on scientific argumentation and communication, with the other half focused on ship design background. The second quarter is focused on ship design. A semester long research assignment examining the influence of alternative energy carriers and converters on the design and operation of a transport vessels will be performed in groups of 2 students. This research assignment will split between a literature study focus in Q1 and a research portion in Q2.	
Study Goals	- Explain and compare primary trade-offs related to alternative power, propulsion, and energy carrier and conversion systems. - Compare and evaluate various aspects of ship design, including: ship systems and ship systems design, and ship layout tradeoffs - Explain the traditional ship design process and its benefits and drawbacks related to sustainable ship design - Search and analyze key challenges and rationale behind sustainable vessel design as part of a literature review - Formulate an argument and communicate key aspects and rationale behind sustainable vessel design in writing as both a literature review and for a specific vessel case study. - Describe and evaluate the influence of environmental maritime considerations on a ship design, as it relates to: ship systems, layout, stability, or ship operations. - Identify and explain primary environmental maritime considerations, which may include aspects of local, national, or international environmental maritime regulations, or vessel retirement recycling	
Education Method	The course will consist of on-campus lectures and a semester long research project in a small group.	
Assessment	Grading will consist of three aspects (1) a group project, (2) scientific argumentation, and (3) an individual assignment	
Department	ME Department Maritime & Transport Technology	

MT2463	Resistance, Propulsion and Powering 2	6
Responsible Instructor	Dr.ir. P. de Vos	
Instructor	E.H.M. Ulijn	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	see Dutch version	
Course Contents	see Dutch version	
Study Goals	see Dutch version	
Education Method	see Dutch version	
Assessment	see Dutch version	
Department	ME Department Maritime & Transport Technology	

MT2464	Mechanics of Maritime Structures 2	6
Responsible Instructor	Dr. C.L. Walters	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Course Contents	Description box (5 ECTS): The course consists of four parts: - Strength of ships: (1) lectures + (2) laboratory exercise - Material science: (3) lectures + (4) laboratory exercise (1) Strength of ships lectures Special topics related to the limit and ultimate strength of ships will be introduced, including buckling (beam and plate), plasticity, fatigue, and fracture.: a) Stress concentration factors b) Buckling of columns: imperfections, eccentricities, Euler, Johnson, Perry-Robertson; c) Buckling of plates: loading, boundary conditions, geometries; d) Effective width of plates: stiffness differences, shear deformations, non-symmetrical/curved cross-sectional forms, post-buckled stress distribution and types of loading; e) Plasticity: form factors, maximum capacity of beams, hinge line theory f) Fracture and fatigue: S-N curves, fatigue crack growth rate, Charpy testing, linear elastic fracture theory, elastic plastic fracture theory. The lecture materials consist of presentations (PowerPoint). The following two reference books are recommended: - Fracture Mechanics: Fundamentals and Applications, ISBN 9781498728133Ship Structural Analysis and Design, ISBN 978-0-939773-78-3 (available for download from TU Delft library)(2) Strength of ships laboratory exercise: plate test (3) Material sciences lectures a) Strengths of materials: characteristics of steel and some non-ferrous ((mechanical) materials, density, structure and properties). b) Strengthening mechanisms on the microstructural scale. c) Thermomechanical processes of metals (deformation, recrystallization, phase transformations). d) Tension test: elastic modulus, yield strength, ultimate tensile strength, failure strain, necking, hardening, and plastic instability. e) Most common welding processes, process descriptions, advantages and disadvantages. f) Influence of welding on materials microstructure, residual stress, deformations, common defects. (4) Material sciences laboratory exercise	
Study Goals	(1) + (2) Strength of ships The recognition of various structural elements (beam and plate elements) of ships with associated supporting conditions and loads. The ability to dimension these structural elements on the basis of strength, stiffness and structural stability. The determination of the stress condition and maximal strength capacity for plate elements loaded in the plane, based on measurements (via digital image correlation). (3) + (4) Materials sciences Insight into the relationship between structure and properties of various materials and into the influence of the production processes on the structure of materials and the use of these materials sciences insights into the design of structures. The ability to describe basic aspects of (fracture-) mechanical material behavior, including the determination of this behavior and the ability to make a simple failure calculation, e.g. for a welded connection. The development of basic knowledge of selected welding processes. Insight into the influence of welding on local microstructures and properties around a welded and the influence of welding on structures. Obtaining insight into the basic principles of melt welding process. Awareness of mechanical tests and the interpretation of results of a tension test.	
Education Method	Lectures and laboratory exercises	
Assessment	a) Laboratory exercise for strength of ships: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. b) Laboratory exercise for material science: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. c) Written exam: this exam consists of a combination of the strength of ships and materials science. 5ECTS	
Department	ME Department Maritime & Transport Technology	

MT3463	Ship Motion and Manouvring	6
Responsible Instructor	Dr.ir. H.J. de Koning Gans	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	ME Department Maritime & Transport Technology	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses.	
Course Contents	WBMT2048 T1 - Analysis 3: - Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss. WBMT2048 T2 - Differential Equations: - Linear first and second order equations; - Laplace transform; - Systems of linear equations - Autonomous systems and stability; - Fourier series; - Partial differential equations.	
Study Goals	Study goals for each part can be found on the corresponding StudyGuide-page (WBMT2048 Toets 1 or WBMT2048 Toets 2).	
Education Method	Lectures 4 hours per week.	
Literature and Study Materials	WBMT2048 Toets 1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that will also be made available via Brightspace. WBMT2048 Toets T2: William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	WBMT2048 T1: - Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the entire course, which lasts 180 minutes. It is not possible to resit only one partial exam. WBMT2048 T2: - Written exams (see Brightspace for more details).	
Exam Hours	WBMT2048 T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048 T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048 T1: A formula sheet (see Brightspace). NO calculators. WBMT2048 T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	
Contact	WBMT2048@tudelft.nl	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master ODE	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-IO

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Dr.ir. G.E. Elsinga	
Education Period	1 2	
Start Education	1	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543	Process Engineering & Thermodynamics	6
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Instructor	Dr. R. Delfos	
Responsible for assignments	L. Botto	
Co-responsible for assignments	Dr.ir. T.M.J. Nijssen	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Modelling Practicals Process Project Small groups Sustainability Technology Thermodynamics Transport phenomena Water Engineering exergy	
Department	ME Department Process & Energy	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633 T2-S	*FEM-practical*	*1*
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-TBM

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Dr.ir. G.E. Elsinga	
Education Period	1 2	
Start Education	1	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543	Process Engineering & Thermodynamics	6
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Instructor	Dr. R. Delfos	
Responsible for assignments	L. Botto	
Co-responsible for assignments	Dr.ir. T.M.J. Nijssen	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Modelling Practicals Process Project Small groups Sustainability Technology Thermodynamics Transport phenomena Water Engineering exergy	
Department	ME Department Process & Energy	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2633 T2-S	*FEM-practical*	*1*
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-TA

CTB2110	Fluid Mechanics	5
Responsible Instructor	Dr. C. Chassagne	
Responsible Instructor	Dr.ir. T.S. van den Bremer	
Instructor	Dr. C. Chassagne	
Instructor	Dr.ir. T.S. van den Bremer	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Collegerama	Yes	

CTB2210	Construction Mechanics 3	5
Responsible Instructor	Prof.dr.ir. J.G. Rots	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Collegerama	Yes	

CTB2300	Dynamics of Systems	3
Responsible Instructor	Dr.ir. K.N. van Dalen	
Responsible Instructor	Dr.ir. H. Hendrikse	
Instructor	Dr.ir. H. Hendrikse	
Co-responsible for assignments	Dr.ir. H. Hendrikse	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Dynamics and Modelling Differential Equations for civil engineering	
Parts	The course consists of lectures in which the course content is provided.	
Summary	To be able to model and analyse the dynamic behaviour of several civil engineering systems	
Course Contents	Mechanical system with one degree of freedom, undamped. Formulate equations of motion for free vibration. Forced vibrations for harmonic, exponential, step and impact/pulse loads. Application of initial conditions. Hydraulic systems with one degree of freem without damping; also free and forced motion. Mechanical system with damping; free and forced vibrations. Hydraulic system with damping; free and forced motion. First order systems as a limit case of 2nd-order systems. Mechanical systems with two degrees of freedom, without damping. Formulat equations of motion (mass matrix, stiffness matrix). Determination of eigenfrequencies and eigenperiods. Forced response for harmonic loads.	
Study Goals	Learning Objectives (LO): LO1: To recognize the properties of structural and hydraulic civil engineering systems (for example mass, elasticity, discharges), that are essential for the modelling of such systems. LO2: To be able to formulate simple equations of motion, using the second Newtons Law, with the help of the displacement method. LO3: To be able to analytically derive solutions for formulated equations of motion. LO4: Interpret solutons with respect to the fundamental dynamics parameters, like: eigenfrequency, eigenperiod, free and forced motion, resonance, and steady-state motion. LO5: To be able to predict descent design improvements for dynamic systems that have been studied. 30 hours lectures 3 hours exam 51 hours individual study and exam preparation --------+ 84 hours (3 ECTS)	
Education Method	Lectures, tutorials	
Course Relations	Dynamics Structural mechanics courses	
Literature and Study Materials	Study guide with general information (BrightSpace) Lecture Notes: "Dynamica van Systemen", J. Blaauwendraad, versie 2010. Lecture Slides Other course materials: "Dynamica van Systemen, Vragenstukkenbundel (oefen- en tentamenopgaven)", Studentassistenten CT2022, versie 2011.	
Assessment	Written exam, with open questions	
Permitted Materials during Tests	Calculator and a sheet with main formulas. The latter will be given at the exam together with the exam questions.	
Expected prior Knowledge	Dynamica & Modelforming; Constructuemechanica 1,2,3; Differentiaalvergelijkingen.	
Academic Skills	Analytical thinking; Critical appraisal.	
Literature & Study Materials	Study guide with general information (BrightSpace) Lecture Notes: "Dynamica van Systemen", J. Blaauwendraad, versie 2010. Lecture Slides Other course materials: "Dynamica van Systemen, Vragenstukkenbundel (oefen- en tentamenopgaven)", Studentassistenten CT2022, versie 2011.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator and a sheet with main formulas. The latter will be given at the exam together with the exam questions.	
Collegerama	Yes	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master RO	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-KT

WB2630 T1 S	Rigid-Body Dynamics	3
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Ir. P.H. de Jong	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Study Goals	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Education Method	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Assessment	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses.	
Course Contents	WBMT2048 T1 - Analysis 3: - Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss. WBMT2048 T2 - Differential Equations: - Linear first and second order equations; - Laplace transform; - Systems of linear equations - Autonomous systems and stability; - Fourier series; - Partial differential equations.	
Study Goals	Study goals for each part can be found on the corresponding StudyGuide-page (WBMT2048 Toets 1 or WBMT2048 Toets 2).	
Education Method	Lectures 4 hours per week.	
Literature and Study Materials	WBMT2048 Toets 1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that will also be made available via Brightspace. WBMT2048 Toets T2: William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	WBMT2048 T1: - Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the entire course, which lasts 180 minutes. It is not possible to resit only one partial exam. WBMT2048 T2: - Written exams (see Brightspace for more details).	
Exam Hours	WBMT2048 T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048 T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048 T1: A formula sheet (see Brightspace). NO calculators. WBMT2048 T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	
Contact	WBMT2048@tudelft.nl	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-CT

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WBMT2048 Toets 1	*Analyse - deeltentamen*	*3*
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Instructor	Drs. A.T. Hensbergen	
Education Period	1	
Start Education	1	
Exam Period	2 1A 1B	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses: Analysis 1 and 2.	
Course Contents	- Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss; - Series and sequences.	
Study Goals	After successfully finishing this course, you are able to: - find parametrisations of curves and surfaces (in 2D and 3D); - calculate a line integral or a surface integral of a function or a vector field; - mention the physical interpretation of these line and surface integrals; - define and formulate the physical meaning of concepts like gradient, curl, and divergence; - calculate line and surface integrals via one of the three integral theorems (Green, Stokes, and Gauss); - determine whether a sequence is convergent or divergent, for example by calculating its limit; - determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test or Ratio Test; - determine the interval of convergence of a power series; - determine the Taylor series of a simple analytic function, for example sin(x), exp(x), ln(x), arctan(x), (1+x)^a.	
Education Method	2 x 2 hour lectures per week.	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	- Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). - Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the complete course, which lasts 180 minutes. It is not possible to resit only one partial exam.	
Exam Hours	Two partial exams of 90 minutes each. One resit of 180 minutes.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	This is half of the course WBMT2048. Partial grades for WBMT2048/T1 (Analysis 3) and WBMT2048/T2 (Differential Equations) will be given in one decimal. The final grade for WBMT2048 is the average of the partial grades for WBMT2048/T1 and WBMT2048/T2. The lowest grade needed for averaging is 5,0. Example: 5,0 for T1 + 7,8 for T2 = 12,8 (Total). So this gives an average of 6,4, your final grade for WBMT2048. The lowest Total for a valid Final Grade (a 'pass') is 11,5 (average 5,75). The best result ever achieved in MyTUDelft/Osiris for a partial grade counts.	
Contact	WBMT2048@tudelft.nl	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-CSE

AESB2210-18	Numerical Mathematics	5
Responsible Instructor	Dr. O. Jaibi	
Co-responsible for assignments	Dr.ir. M. Keijzer	
Contact Hours / Week x/x/x/x	4hrs lecture p/week 4hrs practical p/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This is an introductory course in numerical analysis. Subjects of the course: introduction numerical analysis, interpolation, numerical differentiation, numerical methods for initial value problems, numerical methods for boundary value problems, numerical integration, methods to solve non-linear systems of equations.	
Study Goals	At the end of this course you will be able to 1. interpolate between given data points using: o Lagrange interpolation polynomials; o first order splines; o cubic splines; 2. give upper bounds for the error for a given interpolation method, with and without the effect of rounding/measurement errors; 3. Approximate first and higher derivatives using, but not limited to o central differences; o forward differences; o backward differences; 4. derive a finite difference and show the order of the resulting finite difference; 5. estimate the error using Richardsons order and error estimation; 6. improve a finite difference using Richardsons extrapolation; 7. approximate the solution to nonlinear (systems of) algebraic equations using iterative methods such as o the Picard method; o the bisection method; o the Newton-Raphson method; o the Quasi-Newton method; o the Secant method; o the Regula-Falsi method; 8. select a suitable stopping criteria for a given method; 9. determine whether a given iterative method converges for a given problem and with which order; 10. derive a given iterative method; 11. use numerical integration to approximate integrals, more specifically, using o the (composite) left Rectangle rule; o the (composite) right Rectangle rule; o the (composite) Midpoint rule; o the (composite) Trapezoidal rule; o the (composite) Simpsons rule; 12. give upper bounds for the error for a given numerical integration method, with and without the effect of rounding/measurement errors; 13. solve initial value problems using numerical time-integration methods such as, but not limited to: o the Forward Euler method; o the Backward Euler method; o the Trapezoidal method; o the Modified Euler method; o the Fourth-order Runge-Kutta method; 14. derive the methods given above; 15. derive and analyse the local truncation error for a given numerical time-integration method, with and without use of the test equation; 16. determine or evaluate the stability for a given numerical time-integration method using the amplification factor; 17. choose between two or more given numerical time-integration methods, based on errors, stability and computational effort; 18. estimate the error using Richardsons order and error estimation; 19. solve boundary value problems using the finite difference method; 20. derive and analyse the local truncation error for a given finite difference scheme; 21. determine the stability for a given finite difference scheme using the Gershgorin Circle theorem; 22. choose between central or upwind finite differences; 23. combine a finite difference method with a numerical time-integration method to solve the transient heat equation in one spatial dimension; 24. derive and analyse the local truncation error for a given combination of a finite difference method and a numerical time-integration method; 25. determine the stability bounds for a given combination of a finite difference method and a numerical time-integration method; 26. analyse a given problem; 27. select appropriate numerical methods to solve a given problem; 28. design an algorithm to solve a given problem; 29. implement an algorithm to solve a given problem; 30. visualise the solutions of a given problem; 31. discuss the solutions of a given problem; 32. report on the solutions, the solution algorithm and the used numerical methods for a given problem.	
Education Method	This course consists of lectures and lab work. A timely registration through Brightspace for the lab work is mandatory.	
Computer Use	Lab work to be solved using Python.	
Practical Guide	You are expected to attend the scheduled sessions. Only students in a registered group are allowed to cooperate on the assignments; Notebooks can only be handed in through Assignments within Brightspace. Any notebook handed in any other way will not be assessed; Uploading files into the online practical system Vocareum is strictly prohibited, excluding the Notebook of this academic year of your registered partner for this year; An action of one student counts as an action of the entire registered group; If any of the above rules are violated by a group, that group will be reported to the Board of Examiners for investigation. The violating group will not be notified by us.	
Assessment	The course is assessed by two component examinations: - a written exam (mark, rounded to tenths), which may include a part using assessment software on the students own laptop; - lab work practical containing 3 assignments with reports (V/O). There is also the possibility to do online exercises in the 2nd period, which will result in a mark M, rounded to tenths.	

	In the 2nd period the exam mark E and the mark M obtained in the same period give a final exam mark F using the formula F = max{E,0.85E+0.15M} if E>=5, rounded to tenths. Otherwise and/or in the 3rd period F = E. The lab work reports are graded as V or O, where the lab work result L is a V if all three reports are a V. If F>=5 and L=V, the final mark R of the course is R = F, rounded to halves and wholes.
Tags	Mathematics Numeric Methods Programming
Expected prior Knowledge	- Calculus - Linear Algebra - Differential equations - Python
Academic Skills	Critical and analytical thinking, problem solving, writing a report (for the practical computer assignment)
Literature & Study Materials	Numerical Methods for Differential Equations (C.Vuik, F.J. Vermolen, M.B. van Gijzen, M.J. Vuik): https://textbooks.open.tudelft.nl/textbooks/catalog/book/57
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Non-programmable, non-graphical pocket calculator during on-campus exams
Collegerama	No

WB2630 T1 S	**Rigid-Body Dynamics**	**3**
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Ir. P.H. de Jong	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Study Goals	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Education Method	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Assessment	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Department	ME Department Precision & Microsystems Engineering	

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses.	
Course Contents	WBMT2048 T1 - Analysis 3: - Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss. WBMT2048 T2 - Differential Equations: - Linear first and second order equations; - Laplace transform; - Systems of linear equations - Autonomous systems and stability; - Fourier series; - Partial differential equations.	
Study Goals	Study goals for each part can be found on the corresponding StudyGuide-page (WBMT2048 Toets 1 or WBMT2048 Toets 2).	
Education Method	Lectures 4 hours per week.	
Literature and Study Materials	WBMT2048 Toets 1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that will also be made available via Brightspace. WBMT2048 Toets T2: William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	WBMT2048 T1: - Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the entire course, which lasts 180 minutes. It is not possible to resit only one partial exam. WBMT2048 T2: - Written exams (see Brightspace for more details).	
Exam Hours	WBMT2048 T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048 T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048 T1: A formula sheet (see Brightspace). NO calculators. WBMT2048 T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	
Contact	WBMT2048@tudelft.nl	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-IO

IFEEMCS010400	Linear Algebra	5

Responsible Instructor	Dr. E. Emsiz
Contact Hours / Week x/x/x/x	6/x/x/x
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	Dutch English
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.
Education Method	Colstruction
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.
Permitted Materials during Tests	None
Test and result scale	1-10
Co-Instructor	Dr.ir. R.F. Swarttouw

WB2630 T1 S	Rigid-Body Dynamics	3
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Ir. P.H. de Jong	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Study Goals	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Education Method	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Assessment	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses.	
Course Contents	WBMT2048 T1 - Analysis 3: - Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss. WBMT2048 T2 - Differential Equations: - Linear first and second order equations; - Laplace transform; - Systems of linear equations - Autonomous systems and stability; - Fourier series; - Partial differential equations.	
Study Goals	Study goals for each part can be found on the corresponding StudyGuide-page (WBMT2048 Toets 1 or WBMT2048 Toets 2).	
Education Method	Lectures 4 hours per week.	
Literature and Study Materials	WBMT2048 Toets 1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that will also be made available via Brightspace. WBMT2048 Toets T2: William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	WBMT2048 T1: - Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the entire course, which lasts 180 minutes. It is not possible to resit only one partial exam. WBMT2048 T2: - Written exams (see Brightspace for more details).	
Exam Hours	WBMT2048 T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048 T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048 T1: A formula sheet (see Brightspace). NO calculators. WBMT2048 T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	
Contact	WBMT2048@tudelft.nl	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-MST

WB2630 T1 S	Rigid-Body Dynamics	3
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Ir. P.H. de Jong	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Study Goals	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Education Method	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Assessment	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-NB

AESB2210-18	Numerical Mathematics	5

Responsible Instructor
Dr. O. Jaibi

Co-responsible for assignments
Dr.ir. M. Keijzer

Contact Hours / Week x/x/x/x
4hrs lecture p/week
4hrs practical p/week

Education Period
2

Start Education
2

Exam Period
2
3

Course Language
English

Course Contents
This is an introductory course in numerical analysis. Subjects of the course: introduction numerical analysis, interpolation, numerical differentiation, numerical methods for initial value problems, numerical methods for boundary value problems, numerical integration, methods to solve non-linear systems of equations.

Study Goals
At the end of this course you will be able to
1. interpolate between given data points using:
 o Lagrange interpolation polynomials;
 o first order splines;
 o cubic splines;
2. give upper bounds for the error for a given interpolation method, with and without the effect of rounding/measurement errors;
3. Approximate first and higher derivatives using, but not limited to
 o central differences;
 o forward differences;
 o backward differences;
4. derive a finite difference and show the order of the resulting finite difference;
5. estimate the error using Richardsons order and error estimation;
6. improve a finite difference using Richardsons extrapolation;
7. approximate the solution to nonlinear (systems of) algebraic equations using iterative methods such as
 o the Picard method;
 o the bisection method;
 o the Newton-Raphson method;
 o the Quasi-Newton method;
 o the Secant method;
 o the Regula-Falsi method;
8. select a suitable stopping criteria for a given method;
9. determine whether a given iterative method converges for a given problem and with which order;
10. derive a given iterative method;
11. use numerical integration to approximate integrals, more specifically, using
 o the (composite) left Rectangle rule;
 o the (composite) right Rectangle rule;
 o the (composite) Midpoint rule;
 o the (composite) Trapezoidal rule;
 o the (composite) Simpsons rule;
12. give upper bounds for the error for a given numerical integration method, with and without the effect of rounding/measurement errors;
13. solve initial value problems using numerical time-integration methods such as, but not limited to:
 o the Forward Euler method;
 o the Backward Euler method;
 o the Trapezoidal method;
 o the Modified Euler method;
 o the Fourth-order Runge-Kutta method;
14. derive the methods given above;
15. derive and analyse the local truncation error for a given numerical time-integration method, with and without use of the test equation;
16. determine or evaluate the stability for a given numerical time-integration method using the amplification factor;
17. choose between two or more given numerical time-integration methods, based on errors, stability and computational effort;
18. estimate the error using Richardsons order and error estimation;
19. solve boundary value problems using the finite difference method;
20. derive and analyse the local truncation error for a given finite difference scheme;
21. determine the stability for a given finite difference scheme using the Gershgorin Circle theorem;
22. choose between central or upwind finite differences;
23. combine a finite difference method with a numerical time-integration method to solve the transient heat equation in one spatial dimension;
24. derive and analyse the local truncation error for a given combination of a finite difference method and a numerical time-integration method;
25. determine the stability bounds for a given combination of a finite difference method and a numerical time-integration method;
26. analyse a given problem;
27. select appropriate numerical methods to solve a given problem;
28. design an algorithm to solve a given problem;
29. implement an algorithm to solve a given problem;
30. visualise the solutions of a given problem;
31. discuss the solutions of a given problem;
32. report on the solutions, the solution algorithm and the used numerical methods for a given problem.

Education Method
This course consists of lectures and lab work. A timely registration through Brightspace for the lab work is mandatory.

Computer Use
Lab work to be solved using Python.

Practical Guide
You are expected to attend the scheduled sessions.
Only students in a registered group are allowed to cooperate on the assignments;
Notebooks can only be handed in through Assignments within Brightspace. Any notebook handed in any other way will not be assessed;
Uploading files into the online practical system Vocareum is strictly prohibited, excluding the Notebook of this academic year of your registered partner for this year;
An action of one student counts as an action of the entire registered group;
If any of the above rules are violated by a group, that group will be reported to the Board of Examiners for investigation. The violating group will not be notified by us.

Assessment
The course is assessed by two component examinations:
- a written exam (mark, rounded to tenths), which may include a part using assessment software on the students own laptop;
- lab work practical containing 3 assignments with reports (V/O).
There is also the possibility to do online exercises in the 2nd period, which will result in a mark M, rounded to tenths.

	In the 2nd period the exam mark E and the mark M obtained in the same period give a final exam mark F using the formula F = max{E,0.85E+0.15M} if E>=5, rounded to tenths. Otherwise and/or in the 3rd period F = E. The lab work reports are graded as V or O, where the lab work result L is a V if all three reports are a V. If F>=5 and L=V, the final mark R of the course is R = F, rounded to halves and wholes.
Tags	Mathematics Numeric Methods Programming
Expected prior Knowledge	- Calculus - Linear Algebra - Differential equations - Python
Academic Skills	Critical and analytical thinking, problem solving, writing a report (for the practical computer assignment)
Literature & Study Materials	Numerical Methods for Differential Equations (C.Vuik, F.J. Vermolen, M.B. van Gijzen, M.J. Vuik): https://textbooks.open.tudelft.nl/textbooks/catalog/book/57
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Non-programmable, non-graphical pocket calculator during on-campus exams
Collegerama	No

WB2630 T1 S	**Rigid-Body Dynamics**	**3**
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Ir. P.H. de Jong	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Study Goals	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Education Method	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Assessment	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	**Systems and Control Engineering**	**6**
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-TN

WB2630 T1 S	Rigid-Body Dynamics	3
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Ir. P.H. de Jong	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Study Goals	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Education Method	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Assessment	The code "WB2630 T1 S" should only be used by transfer students ('schakelstudenten') who are *NOT* also taking WB2630 T2. For all other information on the course, see CourseBase "WB2630 Toets 1".	
Department	ME Department Precision & Microsystems Engineering	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master SC	
Contact for students	Schakel-ME@tudelft.nl Toelating-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-CT, B-AES

WB2235	Signal Analysis	6
Responsible Instructor	Dr.ing. R. van de Plas	
Instructor	G. de Albuquerque Gleizer	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	WI2030WBMT Wiskunde 3 (Differentiaalvergelijkingen) WI2031WBMT Wiskunde 4 (Kansrekening en Statistiek) should be heard in parallel to the course	
Course Contents	This course treats the analysis, filtering and detection of signals influenced by linear time-invariant systems as they occur in typical mechanical and mechatronic applications. Signals will be considered in both deterministic and stochastic formulations. Special emphasis is put on discrete-time techniques that can be easily implemented in practice. The main topics of the course are: 1. Elementary properties of signals and systems (even, odd, linear, stable, causal, linear-phase, group-delay, all-pass, minimum-phase) 2. Fourier and Hilbert transform, one and two-sided Laplace and Z-transforms 3. Bode plots and Bodes phase-gain relationship 4. Continuous-time low-pass filters 5. Modulation and demodulation 6. Analog-to-digital and digital-to-analog conversion 7. Design of discrete-time filters using windowing and impulse invariance 8. Elementary properties of random processes (i.i.d., w.s.s., joint w.s.s., independence); key indicators such as mean, auto and cross-correlation/covariance, and spectral densities 9. Identification of linear time-invariant discrete-time systems in the frequency domain 10. Discrete-time Wiener filtering (finite impulse response and non-causal) 11. Detection of discrete-time signals in Gaussian noise	
Study Goals	Students can 1. Evaluate if a signal or system possesses an elementary property; utilize these in simple problems; give examples that exhibit desired elementary properties 2. Compute Fourier, Laplace, z and Hilbert transforms, both directly and using transform pairs and properties 3. Sketch Bode plots for rational transfer functions 4. Determine the specifications of continuous-time low-pass filters from their frequency response; design low-pass filters that fulfill given specifications 5. Analyze typical modulation and demodulation schemes; add missing components 6. Evaluate the impact of A/D and D/A conversion schemes; choose sampling intervals 7. Design discrete-time filters using impulse invariance, windowing, and Wiener techniques 8. Evaluate if a random process possesses an elementary property; utilize them in simple problems; determine key indicators of random processes 9. Identify the frequency response of a linear time-invariant discrete-time system from actual measurements; specify the bias and the variance 10. Derive optimal detection rules for a known signal in Gaussian noise	
Education Method	Lectures, problem sets and practice sessions (werkcolleges).	
Literature and Study Materials	- A.V. Oppenheim, A.S. Willsky and S. Hamid, Signals and Systems, Pearson New International Edition, 2nd ed, 2013. - A.V. Oppenheim and G. Verghese, Signals, Systems and Inference, Class Notes for 6.011: Introduction to Communication, Control and Signal Processing, MIT, Spring 2010. Available online: http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-011-introduction-to-communication-control-and-signal-processing-spring-2010/readings/ - For some parts of the lecture, external materials will be used. These materials will either be available online from within the network of TU Delft, or be provided via Brightspace.	
Assessment	Written exam. Students will only be allowed to bring a non-programmable and non-graphing calculator. A formula sheet will be provided.	
Department	ME Department Delft Center for Systems and Control	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit B-TI, B-IO

WB2235	Signal Analysis	6
Responsible Instructor	Dr.ing. R. van de Plas	
Instructor	G. de Albuquerque Gleizer	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	WI2030WBMT Wiskunde 3 (Differentiaalvergelijkingen) WI2031WBMT Wiskunde 4 (Kansrekening en Statistiek) should be heard in parallel to the course	
Course Contents	This course treats the analysis, filtering and detection of signals influenced by linear time-invariant systems as they occur in typical mechanical and mechatronic applications. Signals will be considered in both deterministic and stochastic formulations. Special emphasis is put on discrete-time techniques that can be easily implemented in practice. The main topics of the course are: 1. Elementary properties of signals and systems (even, odd, linear, stable, causal, linear-phase, group-delay, all-pass, minimum-phase) 2. Fourier and Hilbert transform, one and two-sided Laplace and Z-transforms 3. Bode plots and Bodes phase-gain relationship 4. Continuous-time low-pass filters 5. Modulation and demodulation 6. Analog-to-digital and digital-to-analog conversion 7. Design of discrete-time filters using windowing and impulse invariance 8. Elementary properties of random processes (i.i.d., w.s.s., joint w.s.s., independence); key indicators such as mean, auto and cross-correlation/covariance, and spectral densities 9. Identification of linear time-invariant discrete-time systems in the frequency domain 10. Discrete-time Wiener filtering (finite impulse response and non-causal) 11. Detection of discrete-time signals in Gaussian noise	
Study Goals	Students can 1. Evaluate if a signal or system possesses an elementary property; utilize these in simple problems; give examples that exhibit desired elementary properties 2. Compute Fourier, Laplace, z and Hilbert transforms, both directly and using transform pairs and properties 3. Sketch Bode plots for rational transfer functions 4. Determine the specifications of continuous-time low-pass filters from their frequency response; design low-pass filters that fulfill given specifications 5. Analyze typical modulation and demodulation schemes; add missing components 6. Evaluate the impact of A/D and D/A conversion schemes; choose sampling intervals 7. Design discrete-time filters using impulse invariance, windowing, and Wiener techniques 8. Evaluate if a random process possesses an elementary property; utilize them in simple problems; determine key indicators of random processes 9. Identify the frequency response of a linear time-invariant discrete-time system from actual measurements; specify the bias and the variance 10. Derive optimal detection rules for a known signal in Gaussian noise	
Education Method	Lectures, problem sets and practice sessions (werkcolleges).	
Literature and Study Materials	- A.V. Oppenheim, A.S. Willsky and S. Hamid, Signals and Systems, Pearson New International Edition, 2nd ed, 2013. - A.V. Oppenheim and G. Verghese, Signals, Systems and Inference, Class Notes for 6.011: Introduction to Communication, Control and Signal Processing, MIT, Spring 2010. Available online: http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-011-introduction-to-communication-control-and-signal-processing-spring-2010/readings/ - For some parts of the lecture, external materials will be used. These materials will either be available online from within the network of TU Delft, or be provided via Brightspace.	
Assessment	Written exam. Students will only be allowed to bring a non-programmable and non-graphing calculator. A formula sheet will be provided.	
Department	ME Department Delft Center for Systems and Control	

WB3240-24	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Ir. S.C. Bregman	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Naar Master TM

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit Bewegingswetenschappen / Biomedische Wetenschappen

KT2205	Mathematics 3 and Waves	6.5
Responsible Instructor	Dr. N. Bhattacharya	
Course Coordinator	Dr. K.W.A. van Dongen	
Course Coordinator	Dr.ir. T.M.L. Janssen	
Responsible for assignments	Dr.ir. S. Iskander-Rizk	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2305	Medical Imaging in Pathologies	6.5
Responsible Instructor	Dr. J.F. Veenland	
Course Coordinator	Dr. K.R. Huitema	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2355	Medical Image Processing	3
Responsible Instructor	Dr.ir. M. Staring	
Course Coordinator	Dr. J.F. Veenland	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2705	Designing Medical Technology	3
Responsible Instructor	B. van Vliet	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2755	An Introduction to the Professional Practice	3.5
Responsible Instructor	C. Hallensleben	
Education Period	4B	
Start Education	4B	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2905	Skills and Safety 2	4
Responsible Instructor	Drs. H.G. Buizer-Rijksen	
Course Coordinator	Dr. P.J. Marang-van de Mheen	
Course Coordinator	Drs. I. van Hout	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3405	Intensive Care and Computer Simulation	6.5
Responsible Instructor	P. Somhorst	
Course Coordinator	Dr. A. Schoe	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3505	Complex Diagnostics-Therapy-Combinations	6.5
Responsible Instructor	Dr.ir. J. Schiphof-Godart	
Course Coordinator	Dr. L.M.J.W. van Driel	
Course Coordinator	Prof.dr. M.S. Hoogeman	
Contact Hours / Week x/x/x/x	Gemiddeld per week (10 weken) is: 4u hoorcolleges, 4u opdrachten en practicum, 10u patiëntcasus.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Choose 1 course

KT2505	Senses and Signals	6.5
Responsible Instructor	E. van der Kruk	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2605	Cardiovascular and Respiratory System and Biomedical Instrumentation 2	6.5
Responsible Instructor	S. Man	
Course Coordinator	P. Somhorst	
Course Coordinator	Dr. S.P. Niehof	
Course Coordinator	Prof.dr. P. Steendijk	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2655	Healthcare Information and Organisation	6.5
Responsible Instructor	Dr. R.K. Los	
Responsible Instructor	J. Staal	
Course Coordinator	Dr. S.J. Otto	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit Medische Natuurwetenschappen & Biomedische Technologie

KT2305	Medical Imaging in Pathologies	6.5
Responsible Instructor	Dr. J.F. Veenland	
Course Coordinator	Dr. K.R. Huitema	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2355	Medical Image Processing	3
Responsible Instructor	Dr.ir. M. Staring	
Course Coordinator	Dr. J.F. Veenland	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2505	Senses and Signals	6.5
Responsible Instructor	E. van der Kruk	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2605	Cardiovascular and Respiratory System and Biomedical Instrumentation 2	6.5
Responsible Instructor	S. Man	
Course Coordinator	P. Somhorst	
Course Coordinator	Dr. S.P. Niehof	
Course Coordinator	Prof.dr. P. Steendijk	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2705	Designing Medical Technology	3
Responsible Instructor	B. van Vliet	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2755	An Introduction to the Professional Practice	3.5
Responsible Instructor	C. Hallensleben	
Education Period	4B	
Start Education	4B	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2905	Skills and Safety 2	4
Responsible Instructor	Drs. H.G. Buizer-Rijksen	
Course Coordinator	Dr. P.J. Marang-van de Mheen	
Course Coordinator	Drs. I. van Hout	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2955	Academic Skills 2	4
Responsible Instructor	Dr. N. Aarts	
Course Coordinator	Dr. N. Aarts	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3405	Intensive Care and Computer Simulation	6.5
Responsible Instructor	P. Somhorst	
Course Coordinator	Dr. A. Schoe	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3505	Complex Diagnostics-Therapy-Combinations	6.5
Responsible Instructor	Dr.ir. J. Schiphof-Godart	
Course Coordinator	Dr. L.M.J.W. van Driel	
Course Coordinator	Prof.dr. M.S. Hoogeman	
Contact Hours / Week x/x/x/x	Gemiddeld per week (10 weken) is: 4u hoorcolleges, 4u opdrachten en practicum, 10u patiëntcasus.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit Life, Science and Technology

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijsen	
Responsible for assignments	Dr. R.P. Tas	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Physiology of various cell types (endothelial cells, smooth muscle cells, nurons, and osteoblasts) 3. Musculoskeletal system (bones, joints, musles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 6. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 7. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: Describe the anatomy and the function of the musculoskeletal, nervous, cardiovascular, and respiratory system. Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, from edition 10 on, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 50% of your grade. The (digital) resits are in the Q2 exam period (resit for the Q1 exam) and in the Q3 exam period (resit for the Q2 exam).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

KT2305	Medical Imaging in Pathologies	6.5
Responsible Instructor	Dr. J.F. Veenland	
Course Coordinator	Dr. K.R. Huitema	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2355	Medical Image Processing	3
Responsible Instructor	Dr.ir. M. Staring	
Course Coordinator	Dr. J.F. Veenland	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2505	Senses and Signals	6.5
Responsible Instructor	E. van der Kruk	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2605	Cardiovascular and Respiratory System and Biomedical Instrumentation 2	6.5
Responsible Instructor	S. Man	
Course Coordinator	P. Somhorst	
Course Coordinator	Dr. S.P. Niehof	
Course Coordinator	Prof.dr. P. Steendijk	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2705	Designing Medical Technology	3
Responsible Instructor	B. van Vliet	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2755	An Introduction to the Professional Practice	3.5
Responsible Instructor	C. Hallensleben	
Education Period	4B	
Start Education	4B	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2905	Skills and Safety 2	4
Responsible Instructor	Drs. H.G. Buizer-Rijksen	
Course Coordinator	Dr. P.J. Marang-van de Mheen	
Course Coordinator	Drs. I. van Hout	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2955	Academic Skills 2	4
Responsible Instructor	Dr. N. Aarts	
Course Coordinator	Dr. N. Aarts	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3405	Intensive Care and Computer Simulation	6.5
Responsible Instructor	P. Somhorst	
Course Coordinator	Dr. A. Schoe	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3505	Complex Diagnostics-Therapy-Combinations	6.5
Responsible Instructor	Dr.ir. J. Schiphof-Godart	
Course Coordinator	Dr. L.M.J.W. van Driel	
Course Coordinator	Prof.dr. M.S. Hoogeman	
Contact Hours / Week x/x/x/x	Gemiddeld per week (10 weken) is: 4u hoorcolleges, 4u opdrachten en practicum, 10u patiëntcasus.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bridging Programmes ME**

Vanuit Geneeskunde

KT1005	Mathematics 1 & 2 and Programming	6
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Course Coordinator	P. Somhorst	
Instructor	Dr. N.D. Verhulst	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2305	Medical Imaging in Pathologies	6.5
Responsible Instructor	Dr. J.F. Veenland	
Course Coordinator	Dr. K.R. Huitema	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2355	Medical Image Processing	3
Responsible Instructor	Dr.ir. M. Staring	
Course Coordinator	Dr. J.F. Veenland	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT2605	Cardiovascular and Respiratory System and Biomedical Instrumentation 2	6.5
Responsible Instructor	S. Man	
Course Coordinator	P. Somhorst	
Course Coordinator	Dr. S.P. Niehof	
Course Coordinator	Prof.dr. P. Steendijk	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3405	Intensive Care and Computer Simulation	6.5
Responsible Instructor	P. Somhorst	
Course Coordinator	Dr. A. Schoe	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3505	Complex Diagnostics-Therapy-Combinations	6.5
Responsible Instructor	Dr.ir. J. Schiphof-Godart	
Course Coordinator	Dr. L.M.J.W. van Driel	
Course Coordinator	Prof.dr. M.S. Hoogeman	
Contact Hours / Week x/x/x/x	Gemiddeld per week (10 weken) is: 4u hoorcolleges, 4u opdrachten en practicum, 10u patiëntcasus.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT3806	Clinical Technological Final Project	15
Responsible Instructor	A.J. Loeve	
Course Coordinator	K. Hutchinson	
Education Period	4 3A	
Start Education	3A	
Exam Period	4 5	
Course Language	Dutch	
Department	BSc Clinical Technology (joint-degree)	

KT4001-S	Calculus for Engineering, part 3	1.5
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	KT4001-S is het vak "Calculus for Engineering, deel 3" voor TM schakelstudenten. Zie vakcode "IFEEMCS012300" voor meer informatie over het vak.	

TN2345	Introduction to Waves	3
Responsible Instructor	Dr. K.W.A. van Dongen	
Instructor	Dr. A.A.F.M. Artaud	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	Dutch	
Course Contents	No English version, see Dutch version	
Study Goals	No English version, see Dutch version	
Education Method	No English version, see Dutch version	
Assessment	No English version, see Dutch version	

Dr. N. Aarts

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Department	Biomechanical Engineering

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Dr.ir. J. Bosboom

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Department	Coastal Engineering
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L. Botto

Department	Complexe Vloeistof Processen

Ir. S.C. Bregman

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Drs. H.G. Buizer-Rijksen

Unit	Mech, Maritime & Materials Eng
Department	Biomechanical Engineering

Dr. F.J. van de Bult

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Department	Analysis

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Dr.ir. N.P.K. Chan

Department Statistics

Room 36.HB 06.130

Dr. C. Chassagne

Unit Civiele Techniek & Geowetensch
Department Environmental Fluid Mechanics

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